

SKILL WEEK-4

SESSION-7

Task 1: Process Synchronization and Child Process Monitoring Using waitpid()

```
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP$ mkdir -p skill/week4/task1
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP$ cd skill/week4/task1
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ nano task1.c
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ gcc task1.c -o T1
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ ./T1
Parent Process Started
Parent PID: 947

Parent is waiting for Child PID: 948

Child Process Started
Child PID: 948
Child is working...
Child Process Completed

Child terminated normally
Child Exit Status: 0
Parent Process Completed
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ |
```

Observation: The program successfully creates a child process using `fork()` and uses `waitpid()` to make the parent process wait for the child process to complete. The child process executes its task and terminates normally with exit status 0. After receiving the child's termination status, the parent process resumes and completes successfully. This demonstrates process synchronization and child process monitoring using `waitpid()`.

Task 2: To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

```
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ cd ~/OSSP
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP$ mkdir -p skill/week4/task2
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP$ cd skill/week4/task2
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ pwd
/home/shamikahasini/OSSP/skill/week4/task2
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ nano task2.c
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ ./T2 Madhu
-bash: ./T2: No such file or directory
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ gcc task2.c -o T2
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ ls
T2 task2.c
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ ./T2 Madhu
PATH = /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/gam
es:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/Micros
oftCorporationII.WindowsSubsystemForLinux_2.7.10.0_x64__8wekyb3d8bbwe:/mnt/c
/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/
Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/O
racle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/Sy
stem32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/S
ystem32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:
/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Program Files/
dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Pro
gram Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/shamikahasini/AppData/Local/P
rograms/Python/Python313/Scripts:/mnt/c/Users/shamikahasini/AppData/Local/Progr
ams/Python/Python313:/mnt/c/Users/shamikahasini/AppData/Local/Microsoft/Windows
Apps:/mnt/c/Program Files/Java/jdk-21:/mnt/c/Users/shamikahasini/AppData/Local/P
```

```
PATH = /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/gam
es:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/Micros
oftCorporationII.WindowsSubsystemForLinux_2.7.10.0_x64__8wekyb3d8bbwe:/mnt/c
/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/
Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/O
racle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/Sy
stem32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/S
ystem32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:
/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Program Files/
dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Pro
gram Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/madhulika/AppData/Local/P
rograms/Python/Python313/Scripts:/mnt/c/Users/madhulika/AppData/Local/Progr
ams/Python/Python313:/mnt/c/Users/madhulika/AppData/Local/Microsoft/Windows
Apps:/mnt/c/Program Files/Java/jdk-21:/mnt/c/Users/madhulika/AppData/Local/P
rograms/Microsoft VS Code/bin:/mnt/c/Users/madhulika/AppData/Roaming/npm:/mn
t/c/Users/madhulika/AppData/Local/Microsoft/WinGet/Packages/Gyan.FFmpeg_Micr
osoft.Winget.Source_8wekyb3d8bbwe/ffmpeg-8.1.2-full_build/bin:/snap/bin
```

File found: /usr/bin/pwd

Execute permission: YES

Command resolved successfully.

```
madhulika@LAPTOP-D2M8NKBH:~/OSSP/skill/week4/task2$
```

Observation: The program successfully retrieves the PATH environment variable and searches through its directories to locate the specified command. For the valid command pwd, it finds /usr/bin/pwd and verifies that execute permission is available. For the invalid command Madhu, it correctly reports that the command is not found in PATH. Thus, the program successfully demonstrates PATH parsing, executable searching, permission verification, and handling of missing commands.

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Task 1: To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

```
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ cd ~/OSSP
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP$ mkdir -p skill/week4/task1
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP$ cd skill/week4/task1
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ pwd
/home/shamikahasini/OSSP/skill/week4/task1
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ nano task1.c
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ ./T3
-bash: ./T3: No such file or directory
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ .T1
.T1: command not found
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ gcc task1.c -o T1
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ ./T1
Enter a string: ^C
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ pwd
/home/shamikahasini/OSSP/skill/week4/task1
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ ./T1
Enter a string: Madhu
Original  : Madhu
Expanded  : Madhu
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ ./T1
Enter a string: hello $abc
Warning: Undefined variable: abc
Original  : hello $abc
Expanded  : hello
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task1$ |
```

Observation: The program successfully detects variable references in the input string and expands the variable values using environment variables. For a normal string such as Madhu, the input remains unchanged. When an undefined variable such as \$abc is entered, the program displays a warning and removes the undefined variable from the expanded output. Thus, the program successfully demonstrates variable detection, value expansion, and handling of undefined variables.

Task 2: To Identify Built-ins, Create Dispatch Table, Execute In-Process Built-ins, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ gcc task2.c -o T2
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ ./T2
myshell> exit
Exiting shell...
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ cd ~/OSSP
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP$ cd ~/OSSP/skill/week4/task2
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ pwd
/home/shamikahasini/OSSP/skill/week4/task2
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ nano task2.c
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ gcc task2.c -o T2
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ ls
T2 task2.c
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$ ./T2
myshell> echo Hello World
Hello World
myshell> cd/tmp
Command not found: cd/tmp
myshell> abc
Command not found: abc
myshell> cd /tmp
myshell> pwd
/tmp
myshell> exit
Exiting shell...
shamikahasini@LAPTOP-D2M8NKKBH:~/OSSP/skill/week4/task2$
```

Observation: The program successfully identifies and executes built-in commands such as pwd, echo, cd, and exit using command dispatch logic. The cd command changes the current working directory and the change is maintained when pwd is executed again. Invalid commands are detected and reported as “Command not found.” Thus, the program successfully demonstrates built-in command identification, in-process execution, state maintenance, invalid-command handling, and dispatch logic testing.